

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA0713
NETWORKS

COURSE NAME: COMPUTER

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Antony Magrin J(192572103) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Antony magrin J

Annexure A

ANALYTICAL PROBLEM SOLVING

1. Suppose we want to transmit the message 11100011 and protect it from errors using the CRC polynomial X^3+1 .

- i.) Use polynomial long division to determine the message that should be transmitted.
- ii) Suppose the leftmost bit of the message is inverted due to noise on the transmission link. What is the result of the receiver's CRC calculation? How does the receiver know that an error has occurred?

2. a) A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

b) A network has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communications. The signal- to-noise ratio is 3162. Find the channel capacity

3. A multispecialty hospital uses wireless patient-monitoring devices to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, electromagnetic interference occasionally changes one bit during transmission. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the **Hamming (12,8)** error-correcting code.

4. a) A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

b) A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full

notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

5. A university is conducting an online semester examination for **8,000 students** simultaneously. At exactly **10:00 AM**, the examination server begins transmitting encrypted question papers to all students. The university server is connected through a **10 Gbps high-speed network**, while students access the examination portal using different internet connections such as **fiber broadband (100 Mbps)**, **home Wi-Fi (20 Mbps)**, and **mobile networks (5–10 Mbps)**. During the first few minutes of the examination, many students with slower internet connections report incomplete downloads, repeated retransmissions, delayed acknowledgments, and temporary freezing of the examination portal. Network administrators observe that the server continues transmitting data at a high rate even though several student devices are unable to receive and process packets at the same speed. This results in receiver buffer overflow, unnecessary retransmissions, increased network traffic, and delays in delivering the question papers. **Analyze** the above scenario and identify the primary networking problem. Explain how the mismatch between the sender's transmission speed and the receiver's processing capability affects communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

CSA0713 – Computer Networks: Assessment Tool 1 Answers

Question 1: CRC Error Detection

Given: Message = 11100011, Generator Polynomial = $X^3+1 \rightarrow 1001$ (4 bits)

i) Transmitted Message (Polynomial Long Division)

Since the divisor has 4 bits (degree 3), append **3 zeros** to the message:

Augmented message = **11100011000**

Division (Modulo-2/XOR):

11100011000

÷ 1001

Step	Working Bits	XOR with 1001	Result
1	1110	1001	0111
2	1110	1001	0111
3	1110	1001	0111
4	1111	1001	0110
5	1101	1001	0100
6	1000	1001	0001
7	0010	(no XOR, MSB=0)	0010
8	0100	(no XOR, MSB=0)	0100

Remainder (CRC) = 100

Transmitted Message = Original Message + CRC = 11100011 + 100 = 11100011100

ii) Effect of Inverting the Leftmost Bit

Original transmitted message: **11100011100**

Leftmost bit inverted: $1 \rightarrow 0$

Received message: **01100011100**

At the receiver, this is divided again by 1001 (the generator). Algebraically, the error corresponds to XOR-ing the transmitted message with x^{10} (the leftmost position).

$x^{10} \bmod (x^3+1) = x = 010$ (non-zero)

Result: The receiver's CRC division produces a **non-zero remainder (010)** instead of 0.

How the receiver detects the error: A correctly received frame always yields a **remainder of 0** when divided by the generator polynomial. Since the recalculated remainder here is **010 $\neq$ 0**, the receiver knows the bit was corrupted during transmission and discards/flags the frame for retransmission.

Question 2

a) Throughput Calculation

Given: Bandwidth = 10 Mbps, Frames/minute = 12,000, Bits/frame = 10,000

Frames per second = $12,000 / 60 = 200$ frames/sec

Throughput = Frames per second $\times$ Bits per frame

Throughput = $200 \times 10,000 = 2,000,000$ bps = **2 Mbps**

(Interpretation: Although the network's bandwidth is 10 Mbps, the actual usable throughput is only 2 Mbps.)

b) Channel Capacity (Shannon's Theorem)

Given: Bandwidth (B) = 3000 Hz, SNR = 3162

Formula: $C = B \times \log_2(1 + \text{SNR})$

$C = 3000 \times \log_2(1 + 3162)$

$C = 3000 \times \log_2(3163)$

$C = 3000 \times 11.628$

$C \approx 34,884$ bps ≈ 34.88 kbps

(Interpretation: This is the theoretical maximum error-free data rate achievable on this channel.)

Question 3: Hamming (12,8) Error-Correcting Code

Scenario application: Hospital wireless monitors send 8-bit data (heart rate/BP values) protected using a Hamming(12,8) code so that a single-bit transmission error can be **corrected instantly at the receiver**, without needing retransmission — critical for real-time patient monitoring.

Step 1 – Determine number of parity bits (r):

For 8 data bits ($m = 8$), r must satisfy: $2^r \geq m + r + 1$

- $r = 3 \rightarrow 2^3=8$, but $8+3+1=12 \rightarrow 8 < 12$ (fails)
- $r = 4 \rightarrow 2^4=16$, and $8+4+1=13 \rightarrow 16 \geq 13$ (satisfies) 

So **$r = 4$ parity bits**, giving a total codeword length = $8 + 4 = 12$ bits $\rightarrow$ **Hamming(12,8)**

Step 2 – Bit Positioning:

Parity bits are placed at positions that are powers of 2: **1, 2, 4, 8**

Data bits occupy the remaining positions: **3, 5, 6, 7, 9, 10, 11, 12**

Step 3 – Parity Bit Coverage:

Parity Bit	Position	Covers Bit Positions (where bit position AND parity position $\neq 0$)
P1	1	1,3,5,7,9,11
P2	2	2,3,6,7,10,11
P4	4	4,5,6,7,12
P8	8	8,9,10,11,12

Step 4 – Error Detection & Correction (Receiver side):

The receiver recalculates each parity bit and forms a **syndrome** (binary number) from the check results.

- If syndrome = 0000 → No error.
- If syndrome $\neq$ 0000 → The decimal value of the syndrome directly gives the **exact bit position** that is in error. That bit is flipped (inverted) to correct it automatically.

Relevance to the hospital scenario:

Since interference typically causes only single-bit errors in wireless transmission, Hamming(12,8) allows the monitoring station to **detect and correct the error on the spot**, ensuring doctors receive accurate, real-time patient vitals **without the delay of requesting retransmission** — which is critical in life-monitoring applications.

Question 4

a) IPv4 Subnetting

Given: Network = 172.16.10.0/24, New subnet mask = /27

Step 1 – Subnet bits borrowed: $27 - 24 = 3$ bits

Number of subnets = $2^3 = 8$ subnets

Step 2 – Host bits remaining: $32 - 27 = 5$ bits

Usable hosts per subnet = $2^5 - 2 = 30$ hosts

Step 3 – Block size (increment): $256 - 224 = 32$

List of Subnets:

Subnet	Network Address	Broadcast Address	Usable Host Range
1	172.16.10.0	172.16.10.31	.1 – .30
2	172.16.10.32	172.16.10.63	.33 – .62
3	172.16.10.64	172.16.10.95	.65 – .94
4	172.16.10.96	172.16.10.127	.97 – .126
5	172.16.10.128	172.16.10.159	.129 – .158
6	172.16.10.160	172.16.10.191	.161 – .190
7	172.16.10.192	172.16.10.223	.193 – .222
8	172.16.10.224	172.16.10.255	.225 – .254

IP 172.16.10.78 belongs to: Subnet **172.16.10.64/27** (range 64–95)

- **Network Address** = 172.16.10.64
- **Broadcast Address** = 172.16.10.95

Does 172.16.10.95 fall in the same subnet?

Yes. 172.16.10.95 lies within the range 172.16.10.64 – 172.16.10.95, so it belongs to the same subnet as 172.16.10.78. (Note: .95 is specifically the **broadcast address** of this subnet, not an assignable host address.)

b) IPv6 Addressing

Given address: 2001:0db8:0000:0000:0000:ff00:0042:8329

Step 1 – Compressed (shortest) form:

- Remove leading zeros in each group: 2001:db8:0:0:0:ff00:42:8329
- Replace consecutive all-zero groups with ::

Compressed form = 2001:db8::ff00:42:8329

Step 2 – Expanded (full) form (reversing the compression):

Expanded form = 2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3 – Network Prefix (/64):

First 64 bits (first 4 groups) define the network:

Network Prefix = 2001:0db8:0000:0000::/64 (or compressed: 2001:db8::/64)

Step 4 – Total possible host addresses with /64:

Remaining bits for host portion = $128 - 64 = 64$ bits

Total host addresses = 2^{64}

Question 5: Analysis of the Examination Server Scenario

Primary Networking Problem Identified: Lack of / Failure of Flow Control

Explanation:

The scenario describes a **speed (rate) mismatch between sender and receiver** — the university server transmits at very high speed (10 Gbps) while many student devices/connections (Wi-Fi at 20 Mbps, mobile at 5–10 Mbps) can only receive and process data much slower. This is a classic **flow control problem**, not a bandwidth or congestion problem alone.

How the mismatch affects communication:

1. **Receiver Buffer Overflow:** The server keeps sending frames faster than the slow-connection students' devices can process them. Once a receiver's buffer fills up, any additional incoming data is **dropped**.
2. **Unnecessary Retransmissions:** Since dropped/overflowed packets are never acknowledged, the sender (assuming loss) retransmits them — wasting bandwidth and worsening congestion.
3. **Delayed Acknowledgments:** Overwhelmed receiver devices take longer to process and acknowledge packets, increasing round-trip time and slowing the overall exchange further.
4. **Increased Network Traffic & Freezing:** Repeated retransmissions and unacknowledged data increase overall traffic load, causing timeouts and portal freezing for slower users.

Solution — Flow Control Mechanisms:

This problem is solved by implementing **flow control** protocols such as:

- **Sliding Window Protocol** — limits the number of unacknowledged frames the sender can transmit, matching sender's rate to receiver's processing capability.

- **Stop-and-Wait Protocol** – sender waits for acknowledgment before sending the next frame (simpler, but slower).
- **Dynamic window sizing / rate adaptation** based on receiver's advertised buffer capacity (as used in TCP's flow control via the receive window field).

Conclusion: The core issue is that **the sender's transmission rate exceeds the receiver's processing/consumption rate**, and without proper flow control, this leads to buffer overflow, packet loss, retransmissions, and degraded performance for students on slower connections.